MILITARY SPECIFICATION SHEET

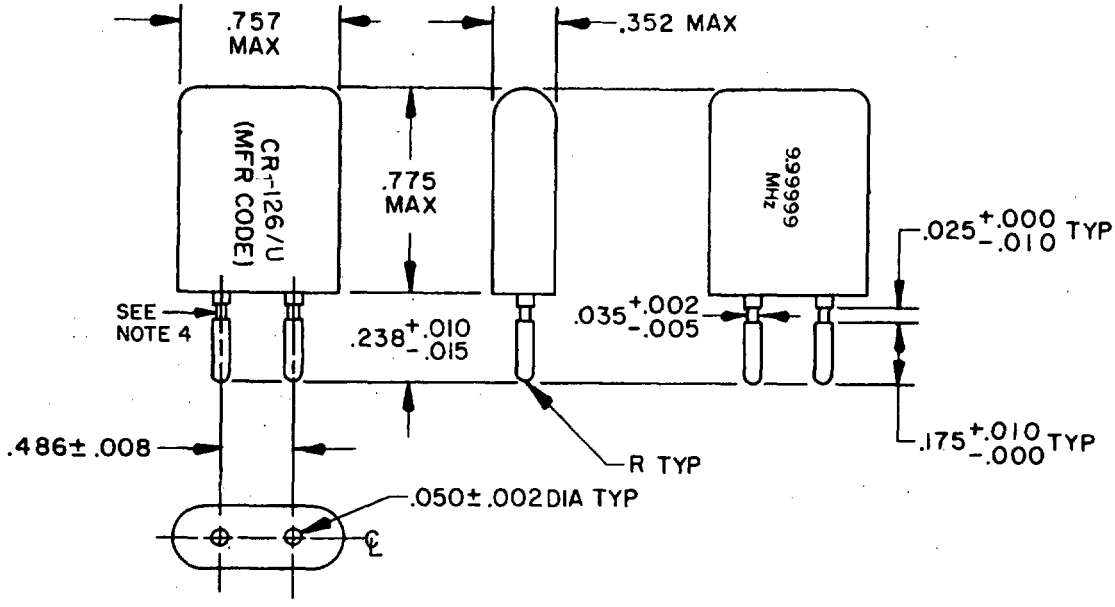
CRYSTAL UNIT, QUARTZ, CR-126/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

(B) INACTIVE FOR NEW DESIGN

Pertinent characteristics: 5 to 10 MHz; third overtone, controlled; antiresonance.



INCHES	MM	INCHES	MM
.002	.05	.050	1.27
.005	.13	.175	4.45
.008	.20	.238	6.05
.010	.25	.352	8.94
.015	.38	.486	12.34
.025	.64	.757	19.23
.035	.89	.775	19.69

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. The type designation, frequency, and code designation characters shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.
4. The pin undercut may be omitted.

FIGURE 1. Crystal unit - CR-126/U.

(B) denotes changes

FSC 5955

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TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-27/U (MIL-H-10056/14)
Dimensions, marking, and configuration	See figure 1
Frequency range	5 to 10 MHz, inclusive
Frequency tolerance	14 ohms maximum
Equivalent resistance	-3.5 ppm +1.5 ppm
Antiresonance, load capacitance	30 ±0.5 pF
Mode of oscillation	Third overtone
Operatable temperature range	-55°C to +90°C, inclusive
Operating	+81°C to 89°C
Test set	TS-330/TSM, modified with RFL AGC adapter model HB-16940 for low drive application
Crystal unit quality (Q) 1/	10^6 minimum
Rated drive level	2 ±1 mW
Capacitance shunt	7 pF, maximum
Shock (specified pulse)	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Thermal shock	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Vibration, high frequency 2/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Temperature run	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent

1/ The crystal unit quality shall be not less than the value specified, as determined during measurement, using the following formula:

$$Q = \frac{1}{4\pi (\Delta f)(C_T)(R)}$$

Where: Δf = difference, in Hz, between the series and antiresonant frequency at C_L (load capacitance) of 30 pF.

C_T = Capacitance: C_0 (shunt capacitance of the crystal unit) plus C_L (load capacitance) of 30 pF

R = Motional-arm resistance of the crystal unit.

2/ Method 204 of MIL-STD-202, test condition A.

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(B) QUALITY ASSURANCE PROVISIONS:

Qualification inspection: Not applicable for this specification.

Quality conformance inspection: Group A tests of MIL-C-3098 shall be applicable.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SM
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 17, 19

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5955-0617-9)

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